

GIANT FIBROEPITHELIAL BREAST LESION MIMICKING PHYLLODES TUMOR: A CASE OF RADIOLOGIC–PATHOLOGIC DISCORDANCE

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BACKGROUND

Fibroepithelial lesions (FELs), particularly fibroadenoma and phyllodes tumor, can demonstrate overlapping clinical, imaging, and histopathologic features. In large or heterogeneous lesions, the diagnostic accuracy of core needle biopsy (CNB) may be affected by sampling limitations, potentially resulting in radiologic–pathologic discordance and diagnostic uncertainty.

OBJECTIVES

- To characterize the multimodality imaging features of a giant fibroepithelial breast lesion.
- To demonstrate the significance of radiologic–pathologic discordance when CNB suggests fibroadenoma despite suspicious imaging features.
- To underscore the role of multidisciplinary assessment in determining appropriate management.

CASE SUMMARY

A 38-year-old nulliparous woman presented with a progressively enlarging right breast mass of approximately 5 years’ duration. Ultrasound demonstrated a giant, multilobulated, hypoechoic solid mass in the lower outer quadrant, with circumscribed margins, non-parallel orientation, and increased intra- and perilesional vascularity. The lesion measured >7 cm and extended beyond the field of view, limiting assessment of the posterior chest wall. Multiple bilateral fibroadenoma-like nodules and simple cysts were also noted. Right axillary lymph nodes were non-reactive, with no suspicious lymphadenopathy. CNB revealed fibroadenoma with myxoid stromal degeneration and stromal hyalinization, without malignancy. Due to radiologic–pathologic discordance, the patient was referred to Dharmais Cancer Center for definitive surgical management.

IMAGING FINDINGS

Ultrasound demonstrated a giant, multilobulated, hypoechoic solid mass in the right breast, with circumscribed margins, non-parallel orientation, and increased intra- and perilesional vascularity. The lesion measured >7 cm and extended beyond the field of view, limiting assessment of the posterior chest wall. Mammography revealed a large dense mass occupying much of the right breast. The lesion was categorized as BI-RADS 4A, with phyllodes tumor and giant fibroadenoma as the main differential diagnoses.

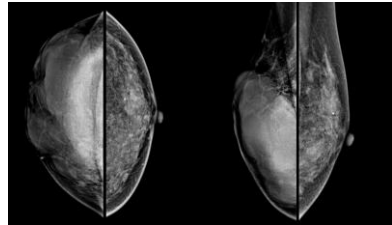


Figure 1. Bilateral mammography: asymmetry with a large dense right breast mass.

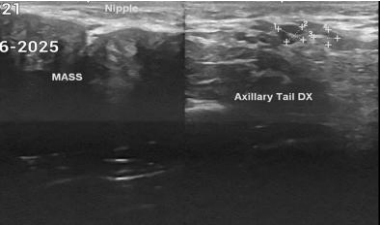


Figure 2. Ultrasound: giant multilobulated mass in the lower outer quadrant.

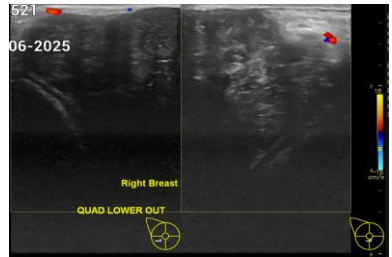


Figure 3. Color Doppler: increased intra-/perilesional vascularity.

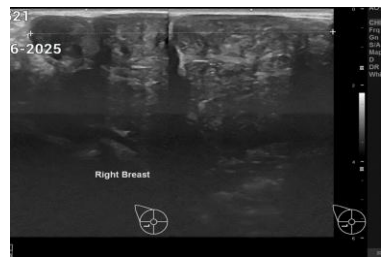


Figure 4. Ultrasound: lesion extends beyond the field of view; size >7 cm.

PATHOLOGY & MANAGEMENT

- CNB:** Fibroadenoma with myxoid stromal degeneration and stromal hyalinization, with no evidence of malignancy.
- RADIOLOGIC–PATHOLOGIC DISCORDANCE:** Despite benign CNB findings, the lesion remained clinically and radiologically suspicious because of its giant size, multilobulated morphology, non-parallel orientation, and increased vascularity.
- MANAGEMENT:** Multidisciplinary evaluation was undertaken, and definitive surgical management was recommended.

TAKE-HOME MESSAGE

1. KEY FINDING

A giant fibroepithelial lesion may radiologically mimic a phyllodes tumor, even when core needle biopsy (CNB) suggests fibroadenoma.

2. CLINICAL IMPLICATION

A benign CNB result may not fully resolve diagnostic uncertainty when imaging findings remain suspicious.

3. PRACTICAL APPROACH

Radiologic–pathologic correlation and multidisciplinary evaluation are essential in the presence of persistent diagnostic discordance.

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